ZHAOXU SUI

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EDUCATION

The Pennsylvania State University 2024 - Present
PhD Candidate (ABD), Geography
Dissertation Title: New Directions for Making Vivid Maps
Advisor: Dr. Anthony C. Robinson
Lab Affiliations: GeoGraphics Lab

University of Oregon 2022 –2024
Master of Science, Geography
Advisor: Dr. Carolyn S. Fish
Thesis Title: Deconstructing Borders, Territories, and Toponyms: Cartographic Designs in the Political Disputed Territory of Sakhalin
Lab Affiliations: Cartofish Lab, S3C Lab, Infographics Lab

University of Minnesota 2019 - 2021
Bachelor of Science, Geography
Minors: GIS, Population Studies, Interdisciplinary Design
GPA: 3.99/4.0, Graduated with High Distinction, Dean's List 2019-2021
Capstone Project: Flooding Risk Assessment in Jinan, China

RESEARCH APPOINTMENTS

Graduate Research Assistant 2025 - Present
Supervisor: Dr. Anthony C. Robinson
Department of Geography, The Pennsylvania State University
Project: Measuring Human Behavior and Ecological Dimensions of Pathogen Transmission for Outbreak Control and Prevention. *National Institutes of Health*. R01TW012434.

Graduate Research Assistant 2023
Supervisor: Dr. Carolyn S. Fish
Department of Geography, University of Oregon
Project: Why Conduct Cartographic Research? A Comparison of Cartographic Research Contributions in Four International Cartographic Association (ICA) Affiliated Journals.

Undergraduate Research Assistant 2021
Supervisor: Jeff Matson
Center for Urban and Regional Affairs (CURA), University of Minnesota
Project: Web Mapping Creative Enterprises in the city of Minneapolis

Undergraduate Research Assistant 2020 – 2021
Institute on the Environment, University of Minnesota
Project: Sweetie Pie – Urban Gardening on Northside Community, Minneapolis

PUBLICATIONS

ORCID 0009-0000-7754-3589

Peer-Reviewed Journal Articles

Sui, Z., Singh, H., & Robinson, A.C. (Submitted). Making vivid maps: Expressive Geovisualization for Representing Spatial Vulnerability. *Cartography and Geographic Information Science Journal.*

Chen, Y., **Sui, Z.**, Deitz, S., Zhou, J., Robinson, A.C., Suarez, M., & Lin, Y. (Submitted). Beyond Map Generation: Cartographic Knowledge, Judgment, and Accountability in the Era of Generative AI. *Annals of the American Association of Geographers.*

Sui, Z., & Fish, C. (2026). Deconstructing Borders, Territories, and Toponyms: Cartographic Designs in the Politically Disputed Territory of Sakhalin. *Cartographic Perspectives*, (110). https://doi.org/10.14714/CP110.2053

Sui, Z. and Lessani, M. N. (2026). Towards Cartographic Design Pluralism: Centering Non-Latin Typographic Practices in Digital Mapping. *GeoHumanties. Outer Cartographies Forum.* https://doi.org/10.1080/2373566X.2026.2631449

Sui, Z., & Robinson, A. C. (2026). Digitally-mediated territorial imaginations: A deep learning approach to characterize online images of contested territories and place names. *Digital Geography and Society*, 100160. https://doi.org/10.1016/j.diggeo.2026.100160

Fish, C., Roth, R., Guidero, E., Raposo, P., **Sui, Z.** (2025). Why Conduct Cartographic Research: A Comparison of Cartographic Research Contributions in Four International Cartographic Association (ICA) Affiliated Journals. *International Journal of Cartography*: 10th Anniversary Issue. https://doi.org/10.1080/23729333.2025.2475411

Peer-Reviewed Extended Abstracts

Robinson, A. C. and **Sui, Z.** (Accepted). Opportunities for Expressive Geovisualization in Mobile Cartography. *Abstracts of 20th International Conference on Location Based Services*. Nanjing, Jiangsu, China. November 11-13, 2026.

Sui, Z. and Robinson, A.C. (2026). Making vivid maps: expressive geovisualization for representing spatial vulnerability. *Abstracts of EuroCarto 2026*. Brno, Czech Republic. September 9-11, 2026. https://doi.org/10.5194/ica-abs-12-137-2026

Sui, Z. (2025). Visualizing States Across Cartographic Designs. *Abstracts of the ICA*. https://doi.org/10.5194/ica-abs-10-277-2025

Sui, Z., & Fish, C. S. (2023). Endonyms or Exonyms: How map purpose, data source, and map language impact place naming on maps. *Abstracts of the ICA*, 6, 1–2. https://doi.org/10.5194/ica-abs-6-249-2023

Other Articles and Book Reviews

Sui, Z. (In Review). When to rotate and when to curve the labels? *Next Generation Cartographers Map Tutorials.*

Sui, Z. (2026). Review of *Body Maps: Improvising Meridians and Nerves in Global Chinese Medicine*. *Cartographic Perspectives, (108).* https://doi.org/10.14714/CP108.2111

Sui, Z., & Lessani, M. N. (2025). Design Principles of Kashida Justification: A Unique Technique in Arabic and Persian Mapping. *Cartographic Perspectives*, (107). https://doi.org/10.14714/CP107.2055

Sui, Z. (2025). Review of *The Rise and Fall of the National Atlas in the Twentieth Century*. *Cartographic Perspectives*, (107). https://doi.org/10.14714/CP107.2013

CONFERENCE PARTICIPATION

Conference Presentations

Sui, Z. (2026). Making vivid maps: Expressive Geovisualization for Representing Spatial Vulnerability. *North American Cartography Information Society Annual Conference, Milwaukee, WI*, October 22, 2026.

Sui, Z. and Robinson, A.C. (2026). Making vivid maps: expressive geovisualization for representing spatial vulnerability. *EuroCarto Conference, Brno, Czech Republic*, September 9, 2026.

Sui, Z. (2026). Expressive Geovisualization: Techniques for Representing Spatial Vulnerability. *American Association of Geographers 2026 Annual Meeting, San Francisco, CA*, March 19, 2026.

Chen, Y. and **Sui, Z.** (2026). Temporal Changes of Static Map Research and Practices in the Digital Era: Evidence from ICA-Affiliated Journals and Exhibitions. *American Association of Geographers 2026 Annual Meeting, San Francisco, CA, March 19, 2026.*

Sui, Z. (2025). Is AI good at reading maps? Envisioning AI-assisted content analysis. *North American Cartography Information Society Annual Conference, Louisville, KY*, October 16, 2025.

Sui, Z. (2025). Characterizing the State across Cartographic Designs. *International Cartographic Conference, Vancouver, British Columbia, Canada*, August 20, 2025.

Sui, Z. (2025). Characterizing State Narratives in Cartographic Designs. *American Association of Geographers (AAG) 2023 Annual Meeting, Detroit, MI*, March 24-28, 2025, & *No Boundaries Symposium, Penn State, State College, PA*, May 2025.

Sui, Z. (2024). Mapping the Transition: The Evolution and Resilience of Specialist Map Retailers around the world. *1st AsiaCarto Conference, Hong Kong, China*, December 11-13, 2024.

Kelso, N., Patterson, T., Plewe, B., **Sui, Z.** (2024). Rebuilding Natural Earth: An Updated Public Preview of Version 6. *North American Cartography Information Society Annual Conference, Tacoma, WA*, October 16-19, 2024.

Sui, Z. (2024). Making a 3D relief globe. *North American Cartography Information Society (NACIS) Annual Conference, Tacoma, WA*, October 16-19, 2024.

Sui, Z. (2024). Deconstructing borders, fill-colors, and toponyms: cartography designs in the political disputed territory of Sakhalin. *American Association of Geographers (AAG) 2024 Annual Meeting, Honolulu, HI*, April 18, 2024.

Sui, Z. (2023). Secure Political Claim: How Japanese Regulate Map Labeling on Southern Sakhalin Island (Karafuto). *North American Cartography Information Society Annual Conference (NACIS), Pittsburgh, PA*, October 11-14, 2023.

Sui, Z. (2023). Endonyms or Exonyms: How map purpose, data source, and map language impact place naming on maps. *International Cartographic Conference, Cape Town, Western Cape, South Africa*, August 14, 2023.

Sui, Z. (2023) AI use in atlas projects. *Map AI Workshop, International Cartographic Conference (ICC), Stellenbosch, Western Cape, South Africa*, August 12, 2023.

Sui, Z. (2023). Endonyms or Exonyms: How map purpose, data source, and map language impact place naming on maps. *American Association of Geographers (AAG) 2023 Annual Meeting, Denver, CO*, March 27, 2023.

Conference Session Organization

Sui, Z. & Mapes, J. (2026). Identity and Community Resilience (Presentation and Panel sessions). *North American Cartography Information Society Annual Conference, Milwaukee, WI*, October 22, 2026.

Sui, Z. and Nawshin, N. T. (2026). Modern Methods in Mapmaking 1: Experimental Cartographies. *American Association of Geographers 2026 Annual Meeting, San Francisco, CA*, March 21, 2026.

Sui, Z. (2025). Research and Design. *North American Cartography Information Society Annual Conference, Louisville, KY*, October 16, 2025.

Houtman, L. and **Sui, Z.** (2024). Putting the Human back in Cartography: Designs and Practices. *American Association of Geographers 2024 Annual Meeting, Honolulu, HI*, April 18, 2024.

FELLOWSHIPS, GRANTS, AND AWARDS

$32,000	University Graduate Fellowship, Penn State Fox Graduate School	2024 - 2026
$222	AAG Cartography Specialty Group Student-Organized Session Funding	2026
$1,200	AAG-International Geographic Information Funds Scholarship Award	2026
$2,500	NACIS Student Travel Grants (Total)	2021 - 2025
$2,000	U.S. National Committee (USNC) to the ICC Travel Funding	2025
€400	International Cartographic Association (ICA) Scholarship for ICC 2025	2025
$75	No Boundaries Symposium Best Presentation Award 2nd Place	2025
$1000	CaGIS Student Scholarship Award	2024
$1000	University of Oregon Rippey Grant Research Awards	2024
$5,000	Anne C. Wilson Graduate Fellowship, Penn State EMS College	2024
$200	AAG Cartography Specialty Group Student-Organized Session Funding	2024
$300	AAG Cartography Specialty Group Masters Student Research Grant	2024
$250	AsiaCarto Conference Travel Grant with Registration Waiver	2024
	AsiaCarto Conference Best Presentation Award	2024
$2,344	U.S. National Committee (USNC) to the ICC Travel Funding	2023
$875	Mather Fellowship, Department of Geography, University of Oregon	2023
$500	International Cartographic Association (ICA) Scholarship for ICC 2023	2023
$25,000	University of Minnesota Global Excellence Scholarship	2019 - 2021
$7,500	University of Minnesota Ecolab Sustainability Scholarship	2020
	Global 10th Place, 15th International Geography Olympiad	2018

SERVICE

College of EMS Graduate Student Council, Penn State
Graduate Council Member 2025 – Present

North American Cartographic Information Society (NACIS)
Co-chair, Diversity, Equity, and Inclusion 2026 – Present
Student Board Member 2024 – 2026
Director, Mentor Program 2024 – 2026

The Graduate Students in the Department of Geography (GSDOG), Penn State
Graduate Representative, Treasurer 2025 – 2026

Belonging, Dignity and Justice (BDJ) Committee, Department of Geography, Penn State
Student Representative 2025

American Association of Geographers (AAG)
Student Chair, Cartography & Mapping Specialty Group 2023 – 2024

Associated Graduate Students in Geography (AGSG), University of Oregon
Colloquium Chair, Representative at Large 2023 – 2024
Social Chair, Treasurer 2022 – 2023

International Student and Scholar Services, University of Minnesota
International Student Ambassador 2020 – 2021

Journal Reviews
Cartographic Perspectives (1), Transactions of GIS (1), Scientific Reports (1)

WORK EXPERIENCE

Graduate Student Cartographer 2024 – Present
GeoGraphics Lab, Penn State

- Commissioned Project: Millbrook Marsh Nature Center Print Companions
- Commissioned Project: Millbrook Marsh Nature Center AR Sandbox Reference Map
- Commissioned Project: Millbrook Marsh Nature Center AR Sandbox Scheme Customization
- Commissioned Project: Millbrook Marsh Nature Center Site Map
- Commissioned Project: Millbrook Marsh Nature Center Waterways Oblique Map
- Commissioned Project: Mapping Repurposed Energy Projects on Disturbed U.S. Lands
- Commissioned Project: A series of maps on Christian Communities in Ming & Qing China
- Commissioned Project: A Reference map of Pennsylvania
- Commissioned Project: Memorial Map of David Myers and the Wildlands Conservancy
  - See media coverage for more information

Commissioned Project: PSU Online Geospatial Education Alumni Map
Commissioned Project: Longitudinal School Attendance Boundary Survey Web Map
Commissioned Project: Millbrook Marsh Nature Center Birding Map
Commissioned Project: Map of French Villages in Mississippi Valley, 17-18th century
Undergraduate Cartographer Mentorship:
Ryan Hillmeyer (Geography)
Julia Byrnes (Geography)

Graduate Student Mapping Assistant 2024
Department of Anthropology, University of Oregon
Project Invisible No More! with Dr. Lynn Stephen
Mapping Mesoamerican language speakers' presence in Oregon, Mexico, and Guatemala.

Graduate Student Cartographer 2022 – 2024
Infographics Lab, University of Oregon
Commissioned Project: Periodic Table of Cartography
Participated Project: Global Initiative of Ungulate Migrations (GIUM) Mapbox Base Map
Participated Project: Global Initiative of Ungulate Migrations (GIUM) Mapbox Animation
Participated Project: USGS Wild Migration Report Maps
Participated Project: Infographics Lab 2022-23 Lab Anthology
Participated Project: GIS Day Event Poster Design

Undergraduate Student Mapping Assistant 2021
Regional Sustainable Development Partnerships, University of Minnesota
Commissioned Project: Lost Creek Hiking Trail with Bluff Country Hiking Club, MN.

TEACHING EXPERIENCE

Teaching Assistantships

Geography 393 & 490: Data Driven Cartography 2023 - 2024
Department of Geography, University of Oregon

Geography 481: GIS I 2023
Department of Geography, University of Oregon

Guest Lectures & Workshops

Designing Reference Maps with ArcGIS Pro and Adobe Illustrator 2026
Invited by Dr. Alicia Iverson for Geography 467: Applied Cartography Design and GeoGraphics Lab, Department of Geography, Penn State

ArcGIS Pro ModelBuilder 2024
Invited by Dr. Nick Kohler for Geography 481: GIS I
Department of Geography, University of Oregon

Cartographic Toponymy 2024
Invited by Dr. Carolyn Fish for Geography 393: Data Driven Cartography.
Department of Geography, University of Oregon

CARTOGRAPHY AWARDS

Arthur Robinson Print Map Award
Cartographic and Geographic Information Society (CaGIS)
Copies are preserved at the Library of Congress, Washington D.C.

"Earth Transit" Honorable Mention 2024
"Travel Map of the World" Honorable Mention 2023
"Comprehensive Map of the World and Beyond" Honorable Mention 2021

Bill Loy Award for Excellence in Cartography
Infographics Lab, Department of Geography, University of Oregon

"Reference Atlas of the World" Honorable Mention 2023

U-Spatial Mapping Prize
Department of Geography, University of Minnesota

"Comprehensive Map of the World and Beyond" Most Ambitious Award 2021

CARTOGRAPHY EXHIBITIONS

Sui, Z. (2026). Alps – An annotated travel map. *North American Cartography Information Society Annual Conference, Milwaukee, WI*, October 21-23, 2026.

Sui, Z. (2026). Central Jinan Architecture & Tourism Map. *North American Cartography Information Society Annual Conference, Milwaukee, WI*, October 21-23, 2026.

Sui, Z., Hillmeyer, R., Coupland., M., & Houtman., L. (2026). Millbrook Marsh Nature Center Site Map. *North American Cartography Information Society Annual Conference, Milwaukee, WI*, October 21-23, 2026.

Sui, Z. (2025). Dynastic Masonry – Visualizing the Great Wall of China. *North American Cartography Information Society Annual Conference, Louisville, KY*, October 15-17, 2025.

Sui, Z. & Byrnes, J. (2025). Penn State Online Geospatial Program Alumni Map. *North American Cartography Information Society Annual Conference, Louisville, KY*, October 15-17, 2025.

Coupland, M. & **Sui, Z.** (2025). A Part of NACIS Map Quilt. *North American Cartography Information Society Annual Conference, Louisville, KY*, October 15-17, 2025.

Sui, Z. (2024). What if China Adopted the US National Park Service System? *North American Cartography Information Society Annual Conference, Tacoma, WA*, October 16-19, 2024.

Sui, Z. (2024). Himalayan Mountainscapes. *British Cartographic Society Map Award Commended Entries*, October 16-19, 2024.

Sui, Z. (2024). Earth Transit. German Newsmedia *ZeitMagazin*, February 2024.

Sui, Z. (2023). Travel Map of the World. *Online Map Gallery of International Cartographic Conference, Cape Town, Western Cape, South Africa, August 14, 2023.*

Sui, Z. (2022). The Sahara. *North American Cartography Information Society Annual Conference, Minneapolis, MN*, October 19-22, 2022.

Sui, Z. (2021). Comprehensive Map of the World and Beyond. *North American Cartography Information Society Annual Conference, Oklahoma City, OK*, October 13-16, 2021.

VOLUNTEER WORK

Data Contributor, *Natural Earth Dataset*	2022 – Present
Graphic Designer, *GISphere*	2023 – Present
User Testing Assistant, *Mapping the Fracking Industry in Western PA*	2026
Cartographer, *Zanjarajuno Natural Reserve, Ecuador*	2024 - 2025
Cartographer, *Department of Anthropology, University of Oregon*	2024
Graduate Workshop Organizer, *Department of Geography, University of Oregon*	2023
Graphic Designer, *Huafeng Magazine Club, University of Oregon*	2022 –2024

SKILLS

GIS	ArcGIS Suite (Pro, Online, Desktop), Mapbox, QGIS
DATA ANALYSIS	Python(Geopandas, ArcPy), Javascript (d3), R Studio, Microsoft
GRAPHIC DESIGN	Adobe Suite (Illustrator, InDesign, Photoshop), Procreate, Plotter Printing
3D MODELING	Bambu Studio, Blender, Cloud Compare (LiDAR)
WEB DESIGN	HTML, CSS, Git/GitHub
LANGUAGES	**Chinese:** Native, **English:** Proficient, **Japanese:** Elementary

MEMBERSHIPS

Cartographic and Geographic Information Society (CaGIS)	2024 – Present
Next Generation Cartographers (NGC), an ICA group	2024 – Present
American Association of Geographers (AAG)	2023 – Present
North American Cartographic Information Society (CaGIS)	2021 – Present
Phi Beta Kappa Society, the University of Minnesota chapter	2021